MAT215: Complex Variables And Laplace Transformations

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LECTURE-04

Normal

Example

- Evaluate
$$\mathscr{L}^{-1}\left\{\frac{1}{s^4}\right\}$$
- Evaluate
$$\mathscr{L}^{-1}\left\{\frac{12}{4-3s}\right\}$$
- Evaluate
$$\mathscr{L}^{-1}\left\{\frac{23s-15}{s^2+5}\right\}$$

First Translation

First Translation Theorem states that:

$$\mathscr{L}^{-1}\{F(s)\} = e^{at}\mathscr{L}^{-1}F(s+a)$$

Example

- Evaluate
$$\mathscr{L}^{-1}\left\{\frac{6}{(s-2)^6}\right\}$$
- Evaluate
$$\mathscr{L}^{-1}\left\{\frac{5}{(s-3)^2+15}\right\}$$
- Evaluate
$$\mathscr{L}^{-1}\left\{\frac{4s}{(s-3)^2+25}\right\}$$

First Translation

Example

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{5s}{s^2+2s+5}\right\}$$

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{6s-4}{s^2-8s-9}\right\}$$

Second Translation

Second Translation Theorem states that:

$$\mathscr{L}\{f(t)u(t-a)\} = e^{-as}\mathscr{L}\{f(t+a)\}$$

$$\mathscr{L}^{-1}\{F(s)e^{-as}\} = f(t-a)u(t-a)$$

Example

- Evaluate
$$\mathscr{L}^{-1}\left\{\frac{1}{s^4}e^{-2s}\right\}$$
- Evaluate
$$\mathscr{L}^{-1}\left\{\frac{1}{(s-3)}e^{-2s}\right\}$$
- Evaluate
$$\mathscr{L}^{-1}\left\{\frac{1}{s^2+4}e^{-\pi s}\right\}$$

Second Translation

Example

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{6s-3}{s^2+4}e^{-\pi s}\right\}$$

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{6s-4}{s^2-8s-9}e^{-\pi s}\right\}$$

Using Partial Fraction Decomposition

Example

- Find the partial fraction decomposition of

$$\frac{1}{s(s+1)}$$

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{1}{s(s+1)}e^{-s}\right\}$$

Hint:

$$\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$$

Using Partial Fraction Decomposition

Example

- Find the partial fraction decomposition of

$$\frac{2s+3}{(s+1)(2s^3+3s^2-3s-2)}$$

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{2s+3}{(s+1)(2s^3+3s^2-3s-2)}\right\}$$

Hint:

$$\frac{2s+3}{(s+1)(2s^3+3s^2-3s-2)} = \frac{1/9}{s+1} + \frac{5/18}{s+2} + \frac{1/2}{s-1} + \frac{-16/9}{2s+1}$$

Using Partial Fraction Decomposition

Example

- Find the partial fraction decomposition of

$$\frac{4s^2 - 5s}{(s+1)(s-2)^2}$$

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{4s^2 - 5s}{(s+1)(s-2)^2}\right\}$$

Hint:

$$\frac{4s^2 - 5s}{(s+1)(s-2)^2} = \frac{1}{s+1} + \frac{3}{s-2} + \frac{2}{(s-2)^2}$$

Using Partial Fraction Decomposition

Example

- Find the partial fraction decomposition of

$$\frac{-s}{(s^2+1)(s+1)}$$

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{-s}{(s^2+1)(s+1)}e^{-\pi s}\right\}$$

Hint:

$$\frac{-s}{(s^2+1)(s+1)} = \frac{-1/2s-1/2}{s^2+1} + \frac{1/2}{s+1}$$

Using Partial Fraction Decomposition

Example

- Find the partial fraction decomposition of

$$\frac{1}{(s^2+1)(s^2+4)}$$

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{1}{(s^2+1)(s^2+4)}\right\}$$

Hint:

$$\frac{1}{(s^2+1)(s^2+4)} = \frac{1/3}{u+1} + \frac{-1/3}{u+4}$$

Using Partial Fraction Decomposition

Example

- Find the partial fraction decomposition of

$$\frac{1}{(s^2+1)(s^2+4)}$$

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{1}{(s^2+1)(s^2+4)}e^{-2\pi s}\right\}$$

Hint:

$$\frac{1}{(s^2+1)(s^2+4)} = \frac{1/3}{s^2+1} + \frac{-1/3}{s^2+4}$$

Using Partial Fraction Decomposition

Example

- Find the partial fraction decomposition of

$$\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}$$

- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}\right\}$$

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- Evaluate

$$\mathscr{L}^{-1}\left\{\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}e^{-3\pi s}\right\}$$

Hint:

$$\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)} = \frac{1/3}{u+2} + \frac{2/3}{u+5}$$